PSET 3: Introduction to Matrices and Vectors

Assignment Qs

- Name
- How long did this problem set take you (round to nearest hour)? 0-1 hour 2-3 hours 4-5 hours 5+ hours
- How difficult was this problem set? very easy 1 2 3 4 5 very challenging

Problem 1: Sketch a function

Sketch the graph of a function (any function you like, no need to specify a functional form) that is:[1]

a. Continuous on $[0, 3]$ and has the following properties: an absolute maximum at 0, an absolute minimum at 3, a local maximum at 1 and a local minimum at 2.

b. Do the same for another function with the following properties: 4 is a **critical point** (i.e. $f'(x) = 0$ or $f'(x)$ is undefined), but there is no local minimum and no local maximum.

Problem 2: Find absolute minimum/maximum values

Find the absolute minimum and absolute maximum values of the functions on the given interval:[2]

a. $f(x) = 3x^2 - 12x + 5$, $[0, 1]$

b. $f(t) = t^2\sqrt{9 - t^2}$, $[-1, 3]$

c. $s(x) = x - \ln(x)$, $[1/2, 2]$

Problem 3: Second derivatives, concavity, and inflection

$$f(x) = \frac{x^3}{15} - x^2 + 4x + 8$$

a. Find $f'(x)$

b. Find $f''(x)$

c. Find $f'''(x)$

[1] inspired by Grimmer HW3.1
[2] inspired by Grimmer HW3.3

d. Find all critical points of f and classify each as a local maximum or local minimum using the second derivative test.

e. Find the inflection point, and state the intervals on which f is concave up and concave down.

Problem 4: Vector arithmetic

Suppose

$$\mathbf{x} = \begin{bmatrix} 3 \\ 2q \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} p+2 \\ -5 \\ 3r \end{bmatrix}$$

If $\mathbf{x} = 2\mathbf{y}$, find p, q, r.[3]

Problem 5: Check for linear dependence

Which of the following sets of vectors are linearly dependent?[4] In each part, you can denote each vector as $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively.

a. $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$

b. $\begin{bmatrix} 13 \\ 7 \\ 9 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ -2 \\ 5 \\ 8 \end{bmatrix}$

c. $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$

Problem 6: Vector length

Find the length of the following vectors:[5]

a. $(4, 2)$

b. $(4, 2, 3, 1)$

c. $(0, 0, 0, 0, 3)$

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[3] inspired by Pemberton and Rau 11.1.3

[4] inspired by Pemberton and Rau 11.1.4

[5] inspired by Simon and Blume 10.10

Problem 7: Documents

Three documents are represented by their counts of the words (*tax*, *vote*, *reform*, *budget*):

$$\textbf{Doc1} = (1, 1, 3, 5) \quad \textbf{Doc2} = (2, 0, 0, 1) \quad \textbf{Doc3} = (3, 0, 1, 2)$$

a. Compute the cosine similarity of Doc1 with Doc2, and of Doc1 with Doc3. Which document is more similar to Doc1?

b. Suppose Doc2 were twice as long, with every word count doubled: $\textbf{Doc2}^* = (4, 0, 0, 2)$. Compute the cosine similarity of Doc1 with $\textbf{Doc2}^*$. Compare to your answer above and explain why this is a desirable property of the measure.

Problem 8: Matrix algebra

Using the matrices below, calculate the following. Some may not be defined; if that is the case, say so.[6]

$$\mathbf{A} = \begin{bmatrix} 3 \\ -2 \\ 9 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 8 \\ 0 \\ -1 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 7 & -1 & 5 \\ 0 & 2 & -4 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 3 & 1 \\ 3 & 4 \\ 3 & -7 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 0 & -4 \\ -2 & 1 & -6 \end{bmatrix}$$

$$\mathbf{F} = \begin{bmatrix} 4 & 1 & -5 \\ 0 & 7 & 7 \\ 2 & -3 & 0 \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} 2 & -8 & -5 \\ -3 & 7 & -4 \\ 1 & 0 & 3 \\ 1 & 2 & 6 \end{bmatrix} \quad \mathbf{K} = \begin{bmatrix} 9 \\ -2 \\ -1 \\ 0 \end{bmatrix} \quad \mathbf{L} = \begin{bmatrix} 5 & 0 & 3 & 1 \end{bmatrix}$$

a. $\mathbf{A} + \mathbf{B}$

b. $\mathbf{A}'\mathbf{B}$

c. $\mathbf{C} + \mathbf{D}$

d. $\mathbf{LK}$

e. $\mathbf{KL}$

f. $\mathbf{E} - 5\mathbf{I}_3$

g. $\mathbf{BC}$

AI and Resources statement

Please list (in detail) all resources you used for this assignment. If you worked with people, list them here as well. It is not enough to say that you used a resource for help, you need to be specific on the link and *how* it was helpful. W/R/T gen AI tools (including GPT, Claude, etc.) you cannot use them to do work on your behalf—you cannot put in any of the questions, etc. You can ask for help on logic / sample problems. If you do use GPT or other AI tools, you need to provide a link to your chat transcript. Any suspected academic integrity violations will be immediately reported to the Dean of Students.

[6]inspired by Grimmer HW5.3